作业 3

3.3. Verify that (3-35) and (3-36) are solutions to (3-34).

$$A_{z1} = C_1 \frac{e^{-jkr}}{r} \tag{3-35}$$

$$A_{z2} = C_2 \frac{e^{+jkr}}{r} \tag{3-36}$$

$$\frac{d^2 A_z(r)}{dr^2} + \frac{2}{r}\frac{dA_z(r)}{dr} + k^2 A_z(r) = 0 \tag{3-34}$$

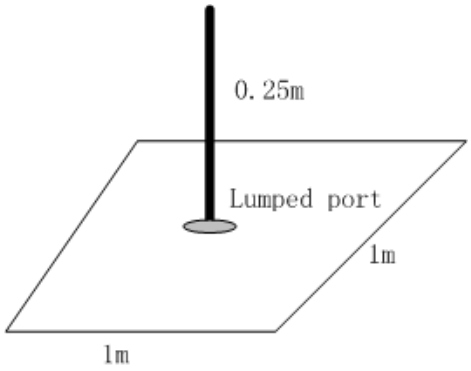
3.6. Derive (3-60) and (3-61).

$$-\nabla \cdot (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) = \mathbf{E}_1 \cdot \mathbf{J}_2 + \mathbf{H}_2 \cdot \mathbf{M}_1 - \mathbf{E}_2 \cdot \mathbf{J}_1 - \mathbf{H}_1 \cdot \mathbf{M}_2 \tag{3-60}$$

第三版第四章问题：4.1, 4.5 (only calculate (a)), 4.29，或第四版第四章问题：4.1, 4.5 (only calculate (a)), 4.31

HFSS Simulation

Frequency: 200MHz-400MHz



To find:

- Zin and VSWR (when this monopole is connected to a 50Ohm transmission line) for the whole frequency band
- Radiation Pattern @300MHz